Lecture 31 : What we have seen so far 1/2

1. Intro week
   - Models aka golems, the statistical engines that process scientific information
   - Think about your justifiable analysis workflow (owls) to drive inferences from data
   - DAGS the launching point to connect scientific knowledge to the topic they study and the inferences made by scientists form data to extract information.
2. Week 2 and 3: Basics of Bayesian inference. Counting, notions of priors and predictions.
3. Week 4: Bayesian linear regression, visualisation, etc. and ore complicated shapes of models (polynomials, splines), but still liner regression at its core.

What we have seen so far 2/2

5. Week 5 and 6: Causal considerations and multiple regression with DAGs: connecting the statistical tools to causal understanding of the system (scientific knowledge) to avoid being confused, biased and confounded.
6. Week 7 and 8: looking at the issue of overfitting and ways to control that tendency. We saw how to study and visualise interactions too.
7. Week 9 and 10: Markov chains and MCMC methods to estimate models without too much assumptions (quadratic approximation) or computational burden (grid). Opens the door to fitting very complex models (even if we won't really see this). E.g. models for the spread of COVID-19 that drove the NZ response in March 2020!

What's next?

Modelling of events

- We are starting to build on the previous tools. Most of the conceptual work is done. We will apply it to new scenarios.
- In particular, we will look at count response events.
- Driving example: University admission data.



Aggregated binomial example

UC Berkeley Admissions

- 4526 graduate school applications for 1973 UC Berkeley
- Stratified by department and gender of applicant
- See `?UCBAdmissions` for details and citations

- Evidence of gender discrimination?

	dept	admit	reject	applications	gender
1	A	512	313	825	male
2	A	89	19	108	female
3	B	353	207	560	male
4	B	17	8	25	female
5	C	120	205	325	male
6	C	202	391	593	female
7	D	138	279	417	male
8	D	131	244	375	female
9	E	53	138	191	male
10	E	94	299	393	female
11	F	22	351	373	male
12	F	24	317	341	female

Before that: maximum entropy principle

Towards Generalised Linear Model 9GLMs)

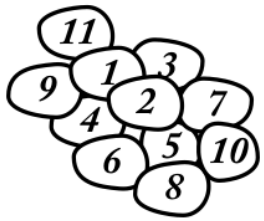
- Choices in statistical models
  - Distributions to represent uncertainty
  - Priors for parameters
  - Likelihood function for data (generation)
- Conventional vs principled choices
- More powerful inference if information is used optimally: all we have, no more.
- Entropy provides a useful principle to guide choices of probability distribution.

Maximum entropy justifications

- The distribution with maximum entropy has the widest and least informative distribution. The uncertainty is spread as evenly as possible, remaining consistent with what we think we know.
- Nature tends to produce empirical distribution that have high entropy. The Gaussian distribution appeared naturally from adding fluctuations (TCL). Other operations/situations create different distributions with maximum entropy.
- It works empirically. There is no guarantee that our small world logic will be useful in the large world of real data. But many difficult problems were successfully solved with maximum entropy choices in the past.

Sorting pebbles into cups



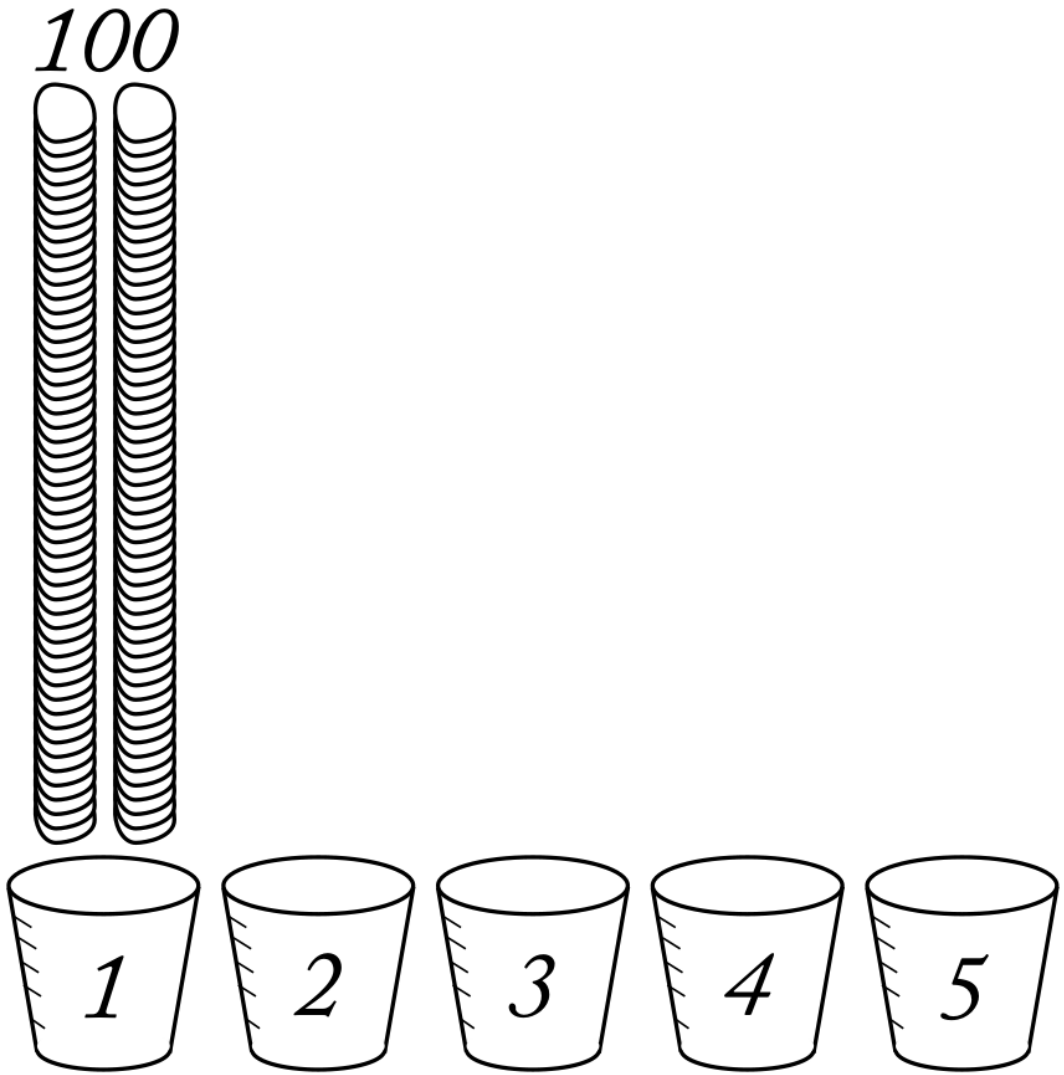


100 pebbles

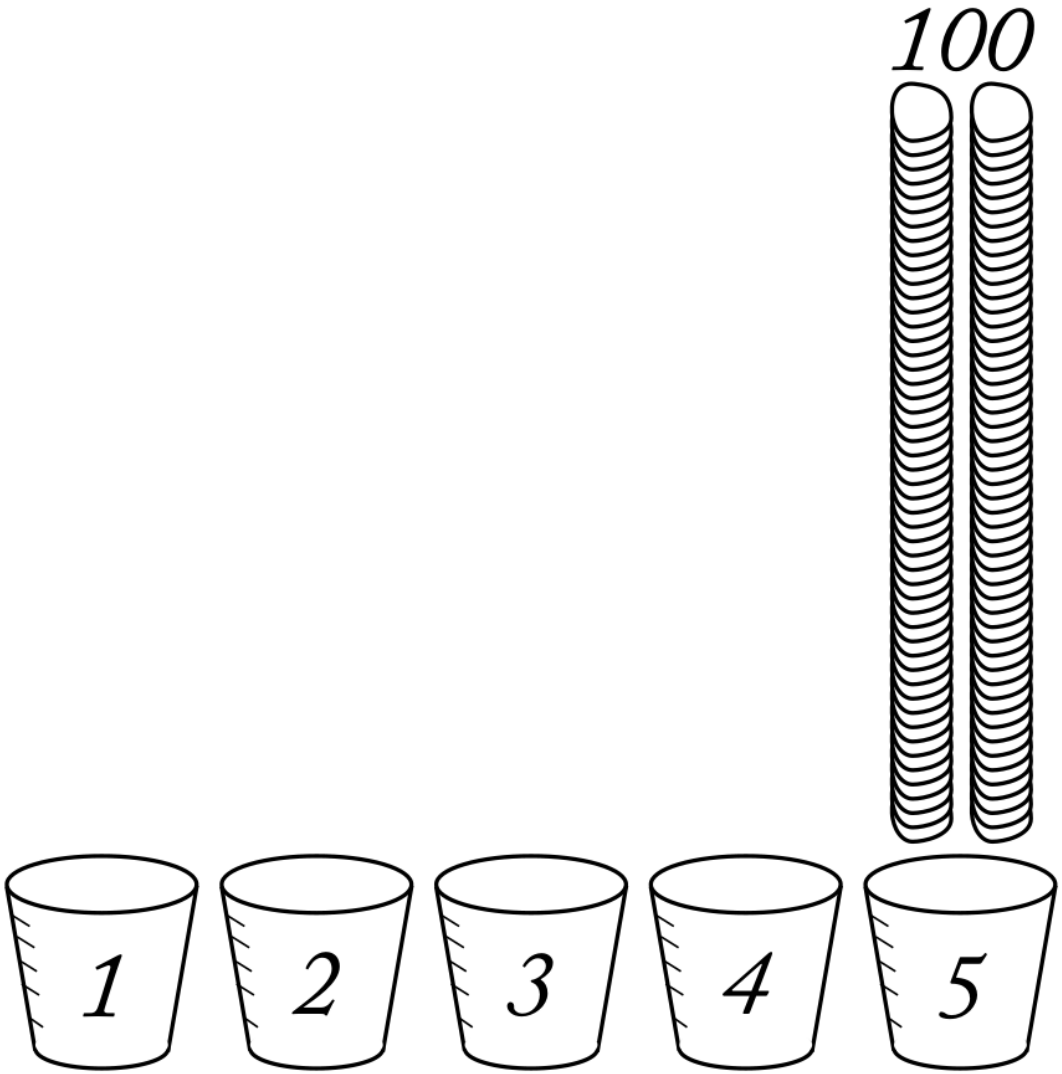
Sorting pebbles into cups



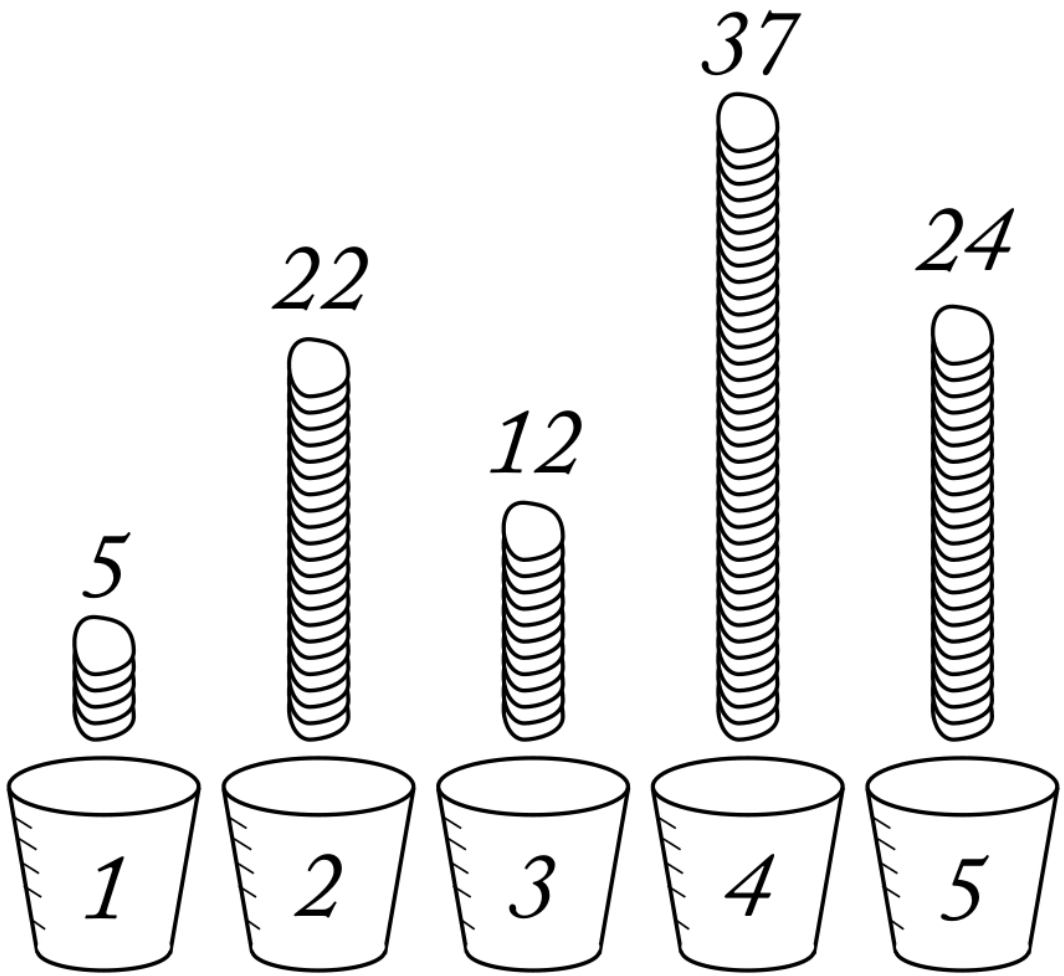
Sorting pebbles into cups



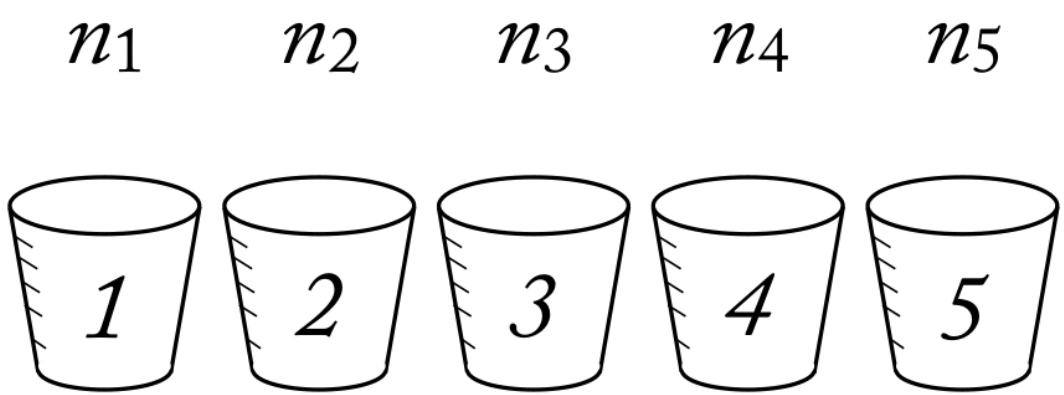
Sorting pebbles into cups



Sorting pebbles into cups



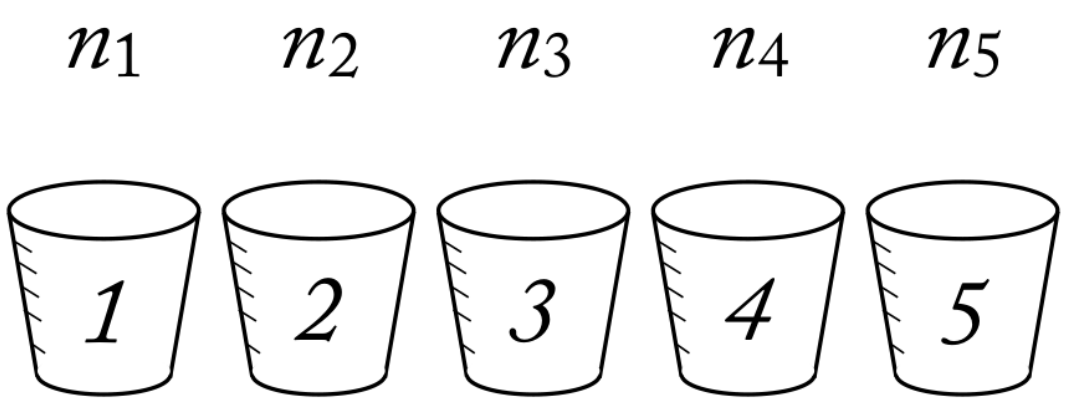
Sorting pebbles into cups



Sorting pebbles into cups

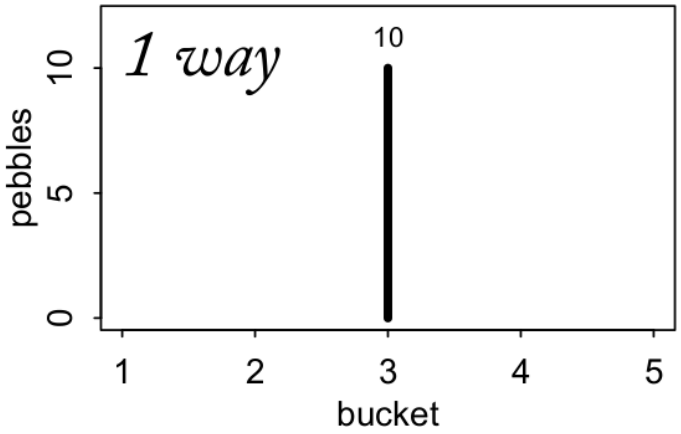
Number of ways:

$$W = \frac{N!}{n_1!n_2!n_3!n_4!n_5!}$$



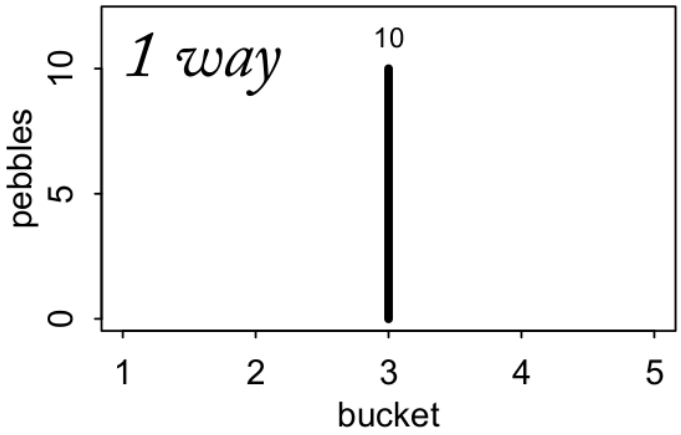
Suppose only 10 pebbles

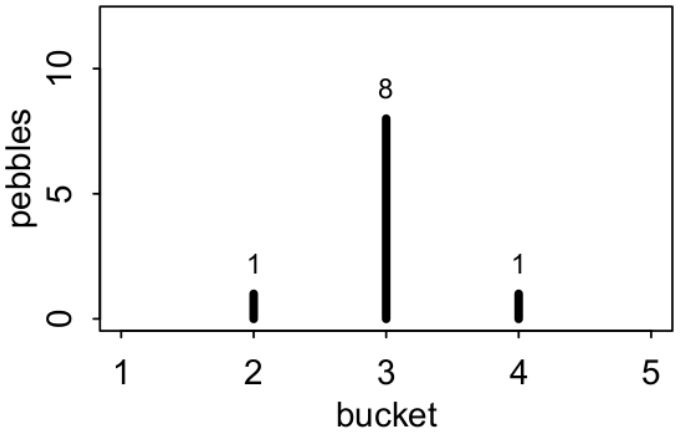
To be able to count



Suppose only 10 pebbles

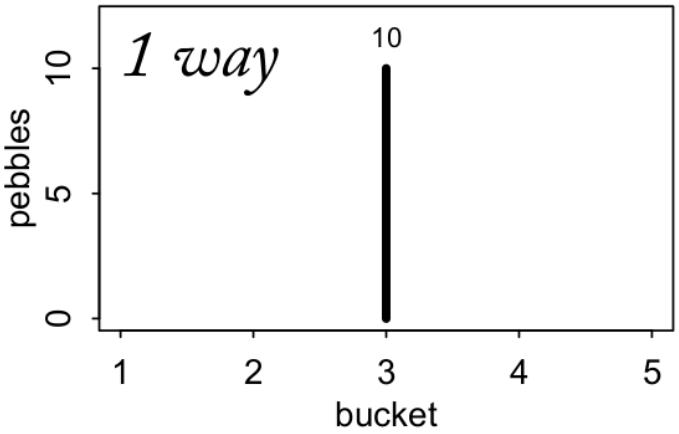
To be able to count

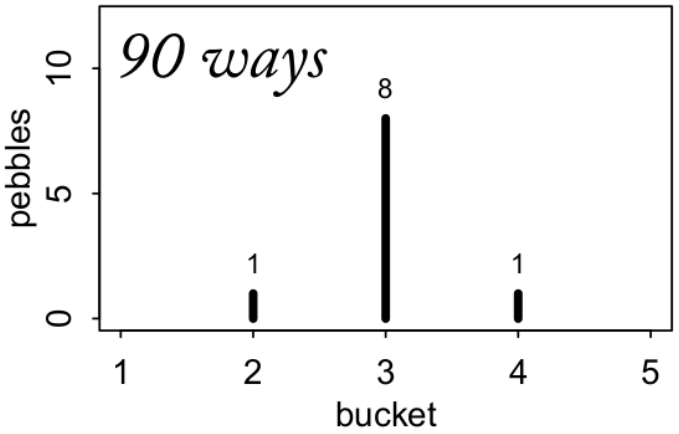




Suppose only 10 pebbles

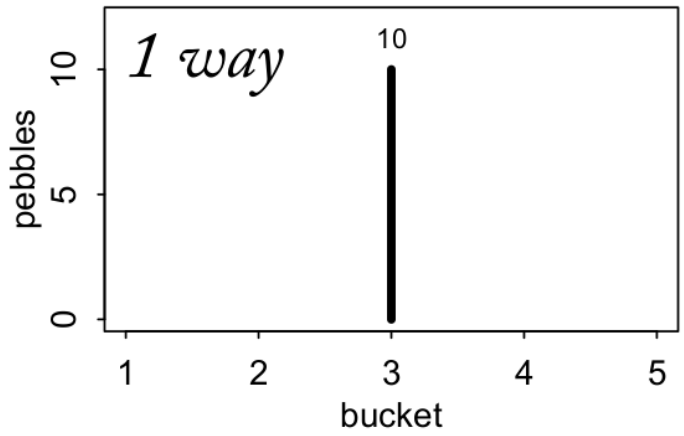
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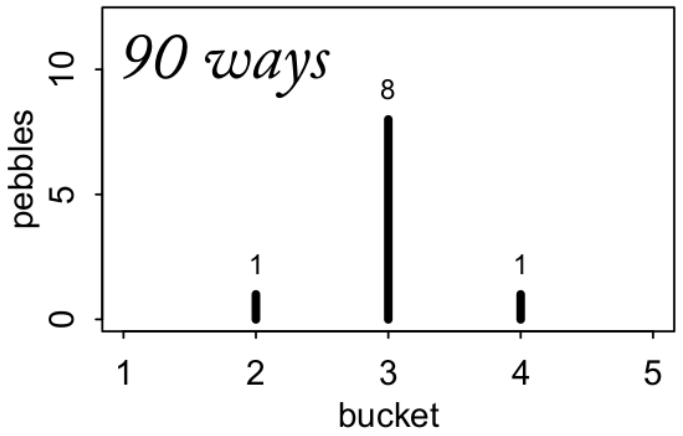


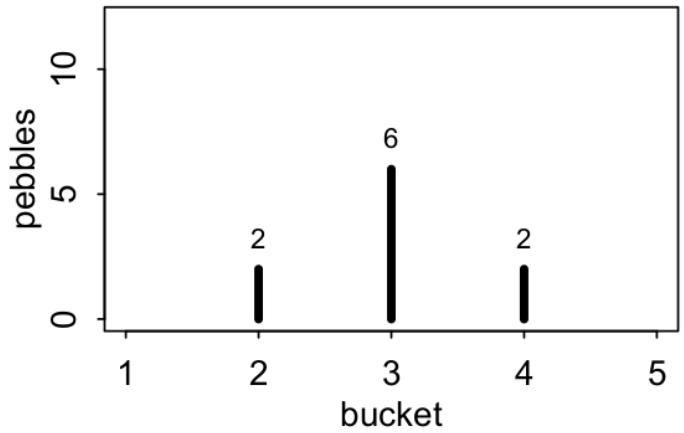


Suppose only 10 pebbles

To be able to count



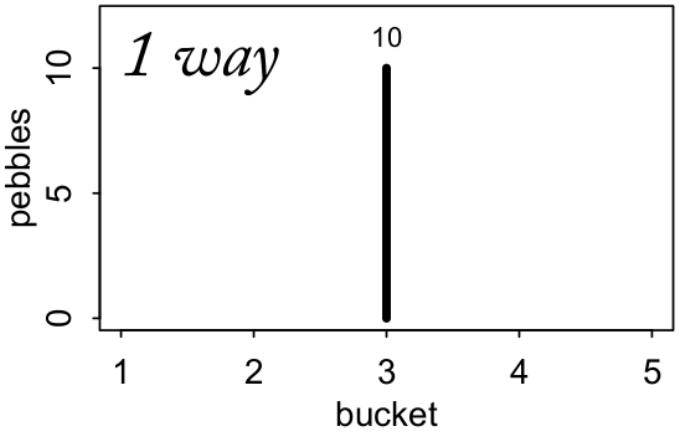


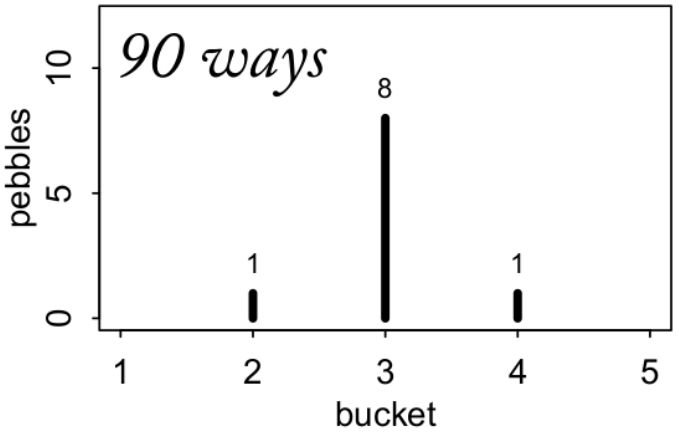


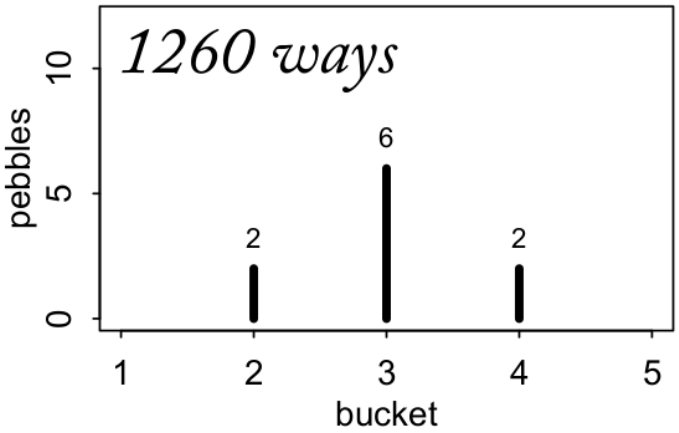
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Suppose only 10 pebbles

To be able to count

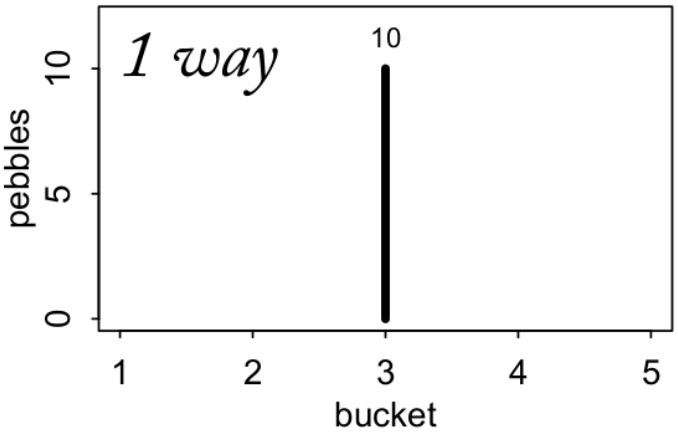


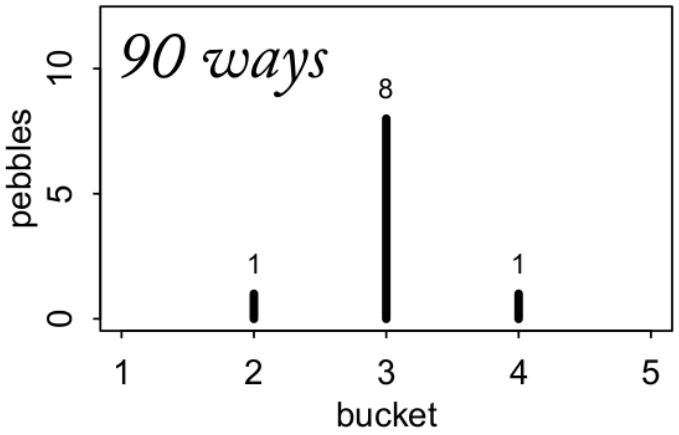


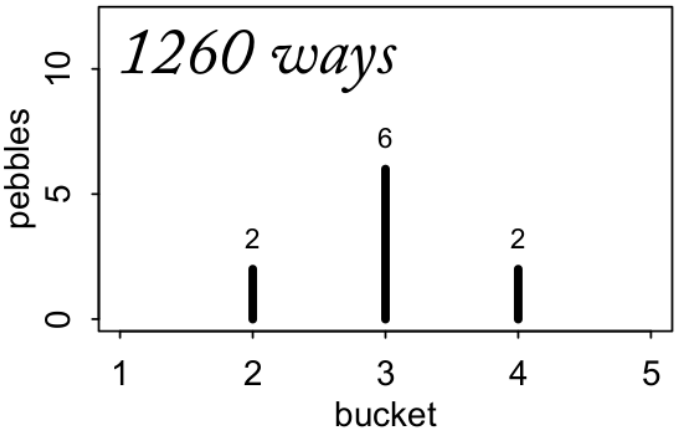


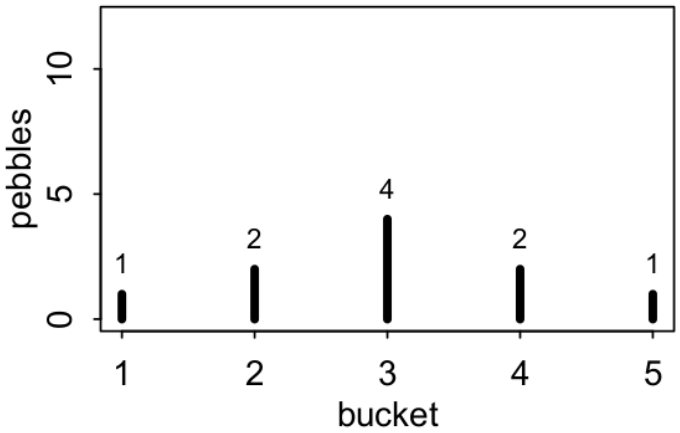
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To be able to count





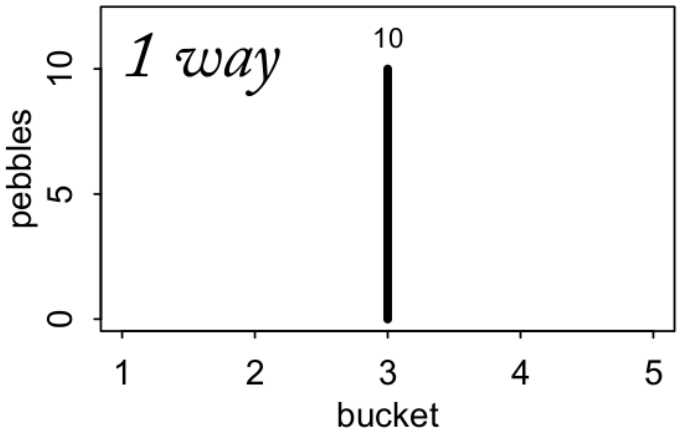


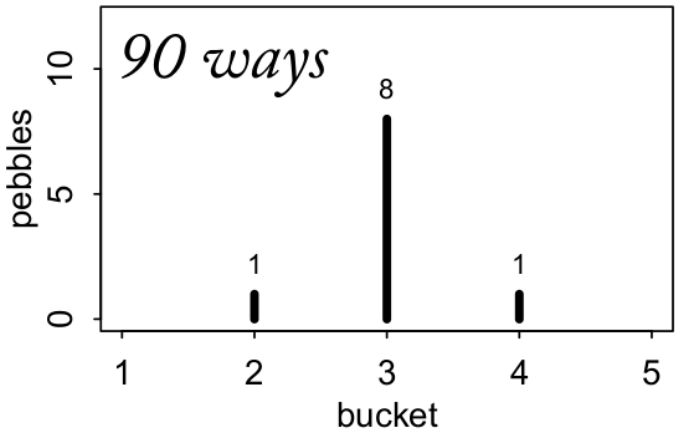


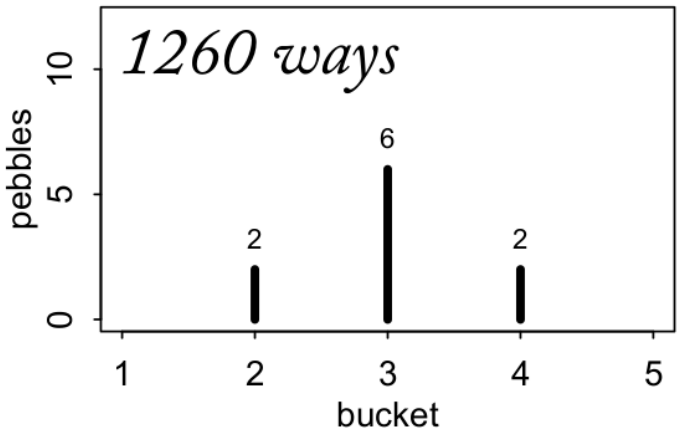
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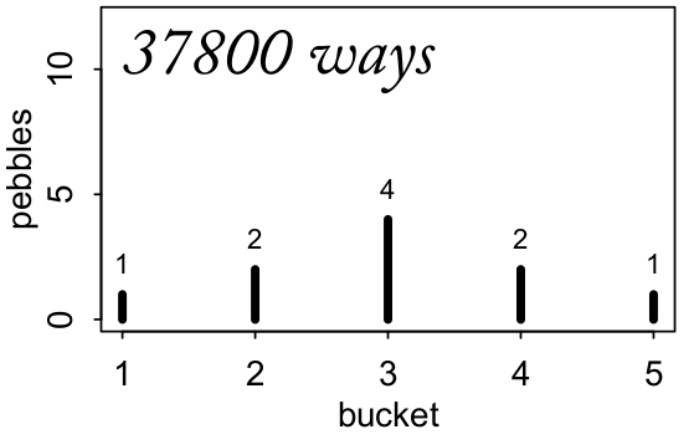
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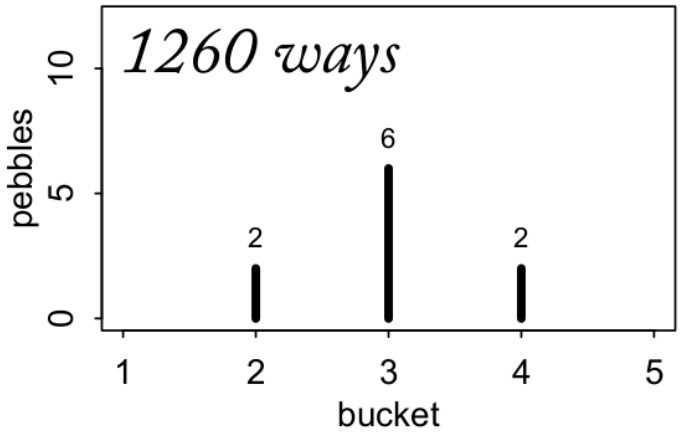


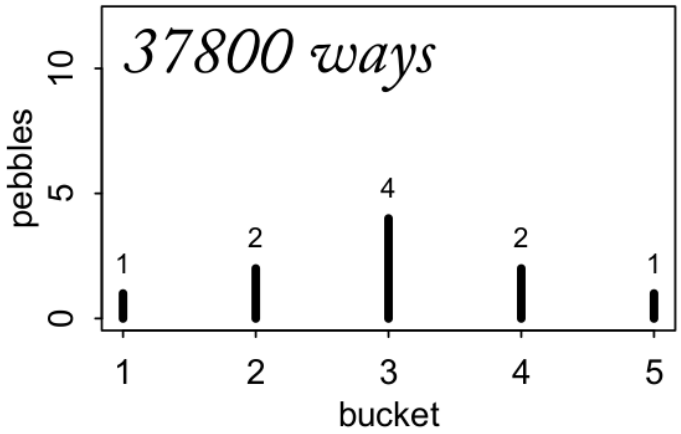


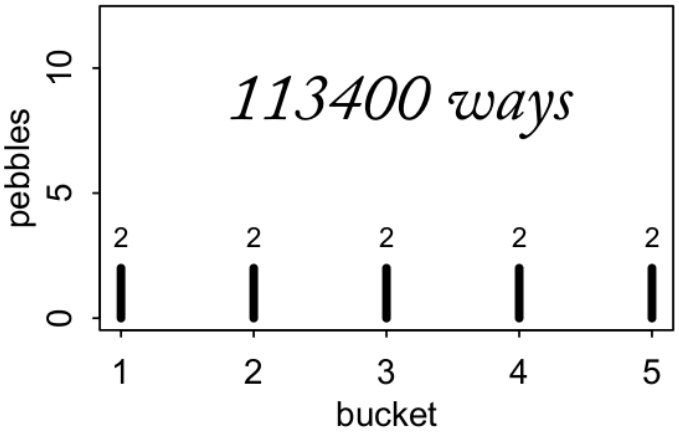
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Suppose only 10 pebbles

The flatter, the more ways



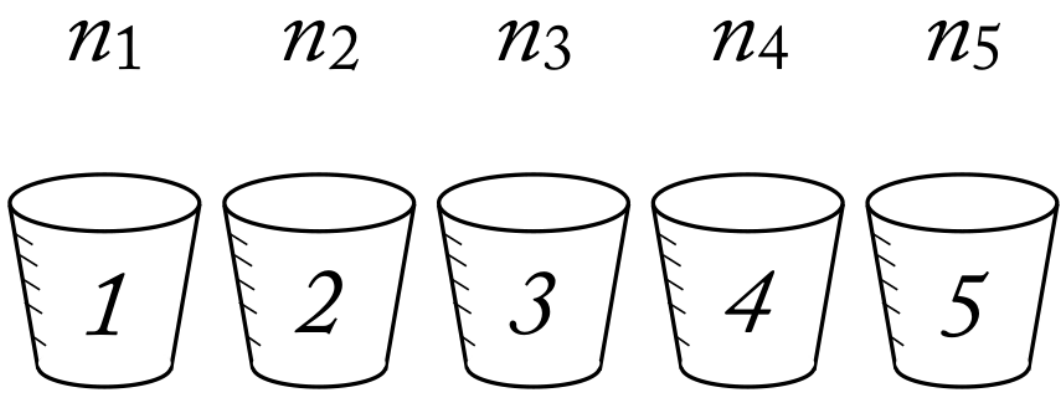




One birth pathway for the entropy

For large N:

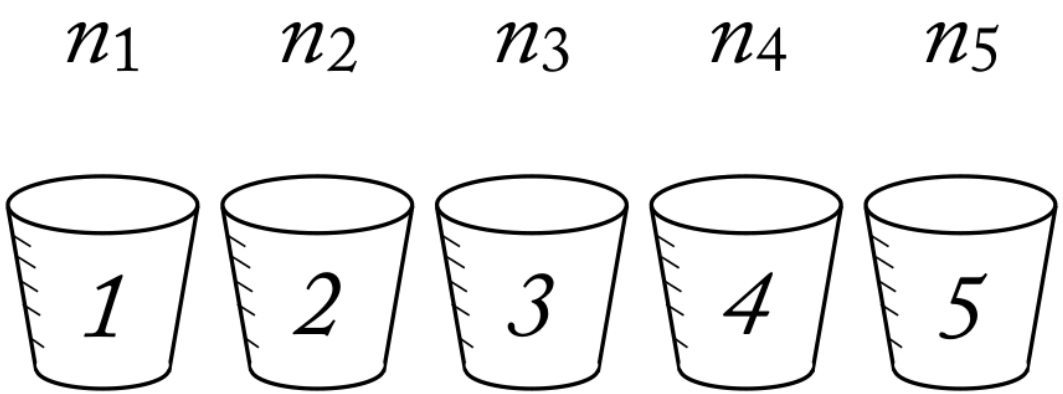
$$\frac{1}{N}\log W \approx -\sum_i \frac{n_i}{N}\log\left(\frac{n_i}{N}\right)$$



One birth pathway for the entropy

For large N:

$$\frac{1}{N}\log W \approx -\sum_i \frac{n_i}{N}\log\left(\frac{n_i}{N}\right) = -\sum_i p_i \log p_i$$



Maximum entropy

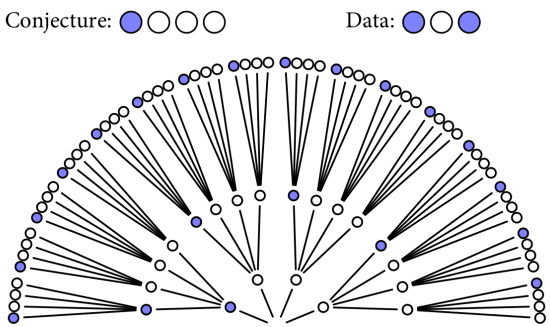
- Previous slide equation matches the information entropy, aka average log-probability (see 10.1 in SR book).
- Maximum entropy principle: the distribution with the largest entropy is the most consistent with the stated assumptions that can happen the largest number of ways
- Way to understand priors for parameters
- Way to understand likelihood for observations (aka data)
- Also explains Bayesian updating as special case (minimum cross-entropy).

Maximum entropy

- Information entropy

$$H(p) = -\sum_i p_i \log p_i$$

- a continuous, additive measure of uncertainty that increases as the possible number of events increases.
- Question: what kind of distribution maximises this quantity?
- Answer: the flattest distribution still consistent with *constraints*.
- Remember the counting example Garden of Forking Data (Section 2.2, SR book): counting paths produces the flattest distribution consistent with stated constraints.



Maximum entropy

Constraints	Maxent distribution
Real value interval	Uniform
Real value, finite variance	Gaussian
Binary events, fixed probability	Binomial
Non-negative real value, has mean	Exponential

Also see the simulations p308-310.

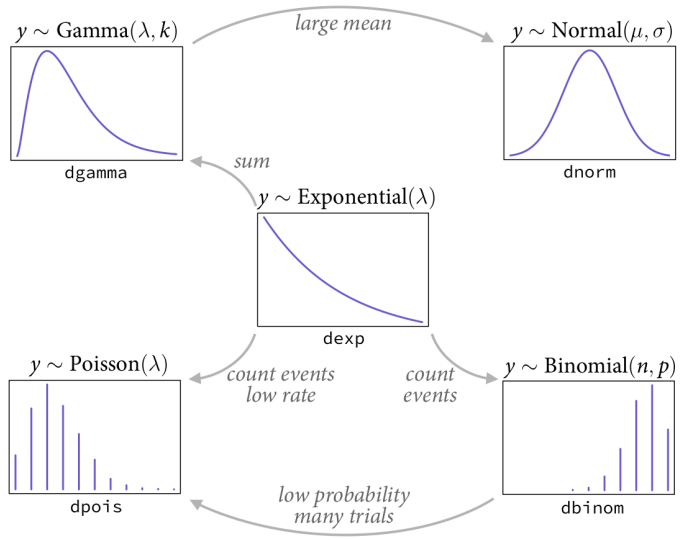
Generalised linear models (GLMs)

- Goal: connect the linear combination of predictors to the outcome variable
- (still geocentric)
- Strategy
  1. Pick an outcome distribution
  2. Model its (transformed) parameters using links to linear models (think about priors)
  3. Compute posterior (for parameters and predictive)
- Can model multivariate relationships and non-linear responses
- Building blocks of multi-level models

Generalised linear models

How to pick a data distribution?

- Mostly from the exponential family
- Arise from natural processes (small set of constraints/principles)
- Maximum entropy interpretation
- Do not pick likelihood by looking at histograms



(see Figure 10.6, SR book)

GLM: data distribution

1. Pick an outcome distribution
   - Distance or duration: exponential, gamma (survival and event history)
   - Counts: Poisson, binomial, multinomial, geometric (e.g. before seeing the data, you know they are non-negative)
   - Not that easy: ranks and ordered categories (glue different sort of link functions; likert scale; etc.), see Chapter 12 of SR book
   - Mixtures: beta-binomial, gamma-Poisson, zero-inflated process, occupancy models...see Chapter 12 of SR book

Generalised linear models

3. Model parameters using a link function f to linear combination of predictors

$$\begin{aligned} y_i &\sim \text{Zaphod}(\theta_i, \psi) \\ f(\theta_i) &= \alpha + \beta.(\mathbf{x_i} - \overline{\mathbf{x}}) \end{aligned}$$

(centering not compulsory, depends on the interpretation)

Notice that the outcome y_i and the parameter θ_i do not need to be of the same nature, on the same domain or in the same unit. Gaussian models were an exception.

Don't forget the priors on the parameter(s)!

GLM: example of the Logit link

3. Model parameters using a link function f to linear combination of predictors

$$\begin{aligned} y_i &\sim \text{Binomial}(n_i, p_i) \\ \text{logit}(p_i) &= \alpha + \beta x_i, \end{aligned}$$

with $\text{logit}(p_i) = \log \frac{p_i}{(1-p_i)}$ (the log-odds) and the inverse-logit is the logistic: $p_i = \frac{\exp(\alpha+\beta x_i)}{1+\exp(\alpha+\beta x_i)}$

Think of the logistic function as a compression of the linear combination of the predictors onto the parameter space.
Other compressions are possible, even for a probability.
Beta prior on p?

See other examples in the SR book.

GLM: posterior computation

4. Compute posterior (for parameters and predictive); you know how to do this.
   - Search space is harder
   - Interpretation of the parameters is harder
   - Link function matters
   - Quadratic approximation can work, but not always. We'll compare `quap()` to `ulam()`.
   - Almost always safer to rely on MCMC.

Enough for today! Tomorrow, we'll see that in GLMs, everything can interact...

Any questions?